

Much of the Imagination About AI and Robots Is Unrealistic

Zhihu · 2025-12-06

Under every question about industrial relocation, complete industrial chains, and the like, one finds views that automation will render physical labour in factories insignificant. Such views are used to argue that 1) China's industries need not relocate, 2) the cheap-labour advantage of underdeveloped countries cannot come into play, 3) and so on.

Since the release of ChatGPT, the capabilities of AI have set people imagining all sorts of things. Many believe AI will throw most white-collar workers out of work. Likewise, autonomous driving and robots will supposedly do for the blue-collar workers. Human labour becomes superfluous; most people have no jobs; in the end, distribution problems must be solved by institutions such as a basic income.

Some even believe the talk of Musk, Altman, and the like that AGI — artificial general intelligence — is about to arrive. Add AGI as the brain to a robot and you have a general-purpose robot. And with all the fancy demonstrations put on by robot companies, many people think the "future has arrived."

Today we will not explore the evolution of capitalist civilisation after the appearance of general-purpose robots — that is a very big question, to be discussed later. This article mainly examines the limits of the current round of AI and robotics technology.

LLM

Because I use large models in my work to automate certain tasks, in actual use I have found the following characteristics of the capabilities of large models:

- 1) They perform especially well on tasks with large token input and small token output: API selection, parameter extraction, task planning based on prompts loaded with background context.
- 2) Large token input and large token output, but with no creative component: for example, translation.
- 3) Small token input, large token output, with no logical requirement: for example, literary and artistic creation.

These tasks can already replace human labour to a considerable degree. But integrating these LLM capabilities into a workflow requires a great deal of customisation, whether in tailoring prompts or fine-tuning models. Moreover, models released after the beginning of this year do not seem to have improved much. No matter how much fine-tuning, optimisation, or engineering is applied, 90% accuracy in real commercial use is already good.

For example, we once considered generating code directly from a requirement description, and we picked a model specifically trained for coding. But in practice, if you do not give it enough background information, the code it writes always lacks this or that; only through endless back-and-forth can you obtain passable code. And "passable" here only means the basic test cases run through — it does not mean bug-free. Anyone who has used it knows AI-written code is riddled with bugs. Products built on this kind of LLM brain can never replace human beings.

What can LLMs replace? Various customer service and process-type white-collar work. Many jobs are in essence extracting parameters from customers or upstream and downstream partners in natural language and filling them into some IT system; replacing this kind of work with an LLM is indeed easy. But that is all. The idea that large models can make most white-collar workers unemployed is impossible.

Several recent reports used to prove the AI bubble have also pointed out this problem.

For example, an MIT report shows that 95% of enterprises fail with generative AI — that is, although many companies are jumping on the AI bandwagon, in practice they cannot use it. There are several similar reports with basically the same conclusion.

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MIT report: 95% of generative AI pilots at companies are failing

 **By Sheryl Estrada**
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August 18, 2025, 6:54 AM ET



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Some may say: won't the capabilities of large models keep improving? Keep stacking GPUs, increase parameter counts, train ever larger models. In fact, the route of "brute force with scaling laws" has hit

its end.

Let me recommend an article written by a Google engineer:

Fundamental Limits of Large Language Model Scaling: Analysis of Mathematical Impossibility Theorems and Resource Constraints

<https://medium.com/hackrlife/fundamental-limits-of-large-language-model-scaling-analysis-of-mathematical-impossibility-theorems-53ece210e23e>

The article collects 25 recent papers, arguing — from the impossibility of hallucination control, data exhaustion, the limits of synthetic data, performance plateaus and diminishing returns, the limits of alignment, and so on — that LLM AI based on the current architecture is what it is, and that further gains through scaling laws will be very hard.

For instance, the following paper discusses how parameter count and returns are not proportional. The relationship between model size and performance gains is logarithmic; each doubling of model size brings less and less benefit.

6.1 Emerging Patterns in Scaling Limitations

Our analysis has revealed several key patterns that characterize the current state of LLM scaling:

- The relationship between model size and performance improvements follows a logarithmic curve, with each doubling of parameters yielding progressively smaller gains. This pattern is particularly evident in benchmark results where GPT-4's 76.60% accuracy on math tasks and Claude 3.5 Sonnet's 90.20% accuracy in code evaluations (Vellum Research Team, 2024) represent significant but increasingly difficult-to-surpass achievements.
- The resource requirements for crossing higher emergence thresholds (θ_c) grow exponentially, while the corresponding performance improvements grow logarithmically. This misalignment between cost and benefit suggests that continued scaling alone cannot be the primary path forward.
- Data quality and diversity have emerged as potentially more critical constraints than raw computational power. The variance term $V(P, D)$ in our bias-variance decomposition increasingly dominates performance limitations, indicating that data innovation may be more crucial than model size scaling.

Has LLM Reached the Scaling Ceiling Yet? Unified Insights into LLM Regularities and Constraints

<https://arxiv.org/pdf/2412.16443>

The following paper proves that hallucination is unavoidable; eliminating hallucination in LLMs is impossible. This is not a matter of LLM architecture but of information aggregation, and is therefore inevitable.

9 Summary

We have established that no large language model can simultaneously achieve perfect hallucination (or imagination) control. We showed that four essential properties—truthful knowledge representation, semantic information conservation, complete revelation of relevant knowledge, and optimal response generation—are mutually incompatible. This impossibility emerges from the mathematical structure of information aggregation itself, not from limitations in data, compute, or architecture.

We introduced a rigorous mathematical framework and grounded in it the general analysis of inference processes and their LLM implementations. At the heart of our mathematical framework lies the semantic information measure μ_C and the emergence operator $\mathcal{E}_C$. The earlier one is a context-dependent metric capturing how knowledge reduces uncertainty within computational bounds. The later one formalizes how reasoning makes latent knowledge explicit rather than creating it anew. Finally, we model LLM inference as an auction of ideas—a idealized marketplace where neural components compete by bidding with their partial knowledge, each trying to influence the final response.

The impossibility manifests through three complementary proofs, each illuminating different aspects of the fundamental constraint. Through game theory, we apply the Green-Laffont theorem to show the impossibility when knowledge components are independent. Through probability theory, we leverage Savage's proper scoring rules to extend the result to general trained or fine-tuned probability distributions. And through direct analysis of transformer architectures, we demonstrate how log-sum-exp convexity in attention mechanisms creates measurable violations of information conservation. The last proof exploits Jensen inequality for convex mappings to define the precise mathematical quantity that measures how aggregation of ideas (auction bids) produces overconfident responses that the constituent components cannot justify with revealed knowledge.

These mathematical insights we have made yield some interesting philosophical implications. Hallucination and creativity (two sides of the same coin we call imagination) emerge as mathematically identical phenomena, distinguished only by our normative judgments. The impossibility result explores the outer limits of reason, identified by Gödel's incompleteness in mathematics, Heisenberg's uncertainty in physics, and Arrow's impossibility in social choice. Also, our framework seems to provide mathematical foundations and experimental settings for models of artificial consciousness. That may open interesting avenues for future research.

On the Fundamental Impossibility of Hallucination Control in Large Language Models <https://arxiv.org/pdf/2506.06382>

Interested friends may read the other papers themselves.

Coincidentally, Ilya Sutskever, one of the GPT authors, has recently been saying that the era of scaling is over, and it is time for research labs to study new architectures. In other words, this round of AI commercialisation is over; we must go back to the lab and wait for the next technological breakthrough to open a new round of commercial boom. He may have his own motives for saying this, but people in the industry have long said the same. The main point is that he is the proponent of the scaling law, so his pronouncement of its death carries authority.

Ilya Sutskever Declares the Scaling Era Dead. His \$3 Billion Bet Says Research Will Win.

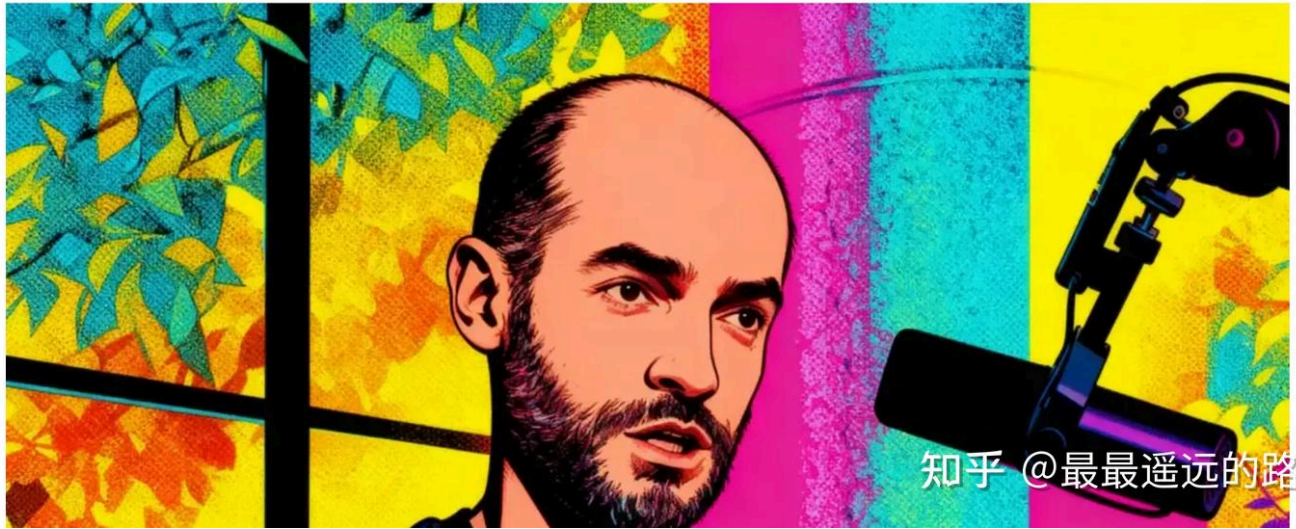
Ilya Sutskever helped prove that scale works in AI. Now he says it doesn't—and his \$3 billion company is betting the future belongs to researchers with ideas, not labs with the biggest GPU clusters. The industry isn't ready for this argument.



MARCUS SCHULER

NOVEMBER 25 2025 • 10:20 AM (UTC-08:00 • PST) • 10 MIN READ

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Robots

The hottest claim about robots today is that humanoid robots, being able to adapt to the whole infrastructure built around human workers, can smoothly replace human labour. Once the humanoid robot reaches its own "ChatGPT moment," billions of manual labourers around the world can be replaced — a gigantic market.

I myself am not in robotics and have no relevant experience. But I have always been sceptical of the recent claims that robots will make assembly-line workers superfluous.

Robot factories could threaten jobs of millions of garment workers



📷 Workers on a production line in Ho Chi Minh City, Vietnam. Nine million people, mostly young women, are dependent on garment and footwear jobs in south-east Asia. Photograph: Richard Vogel/AP

As Adidas announces a new robot shoe factory, an ILO report says up to 90% of workers in south-east Asia could face unemployment due to automation

The jobs of nearly 90% of garment and footwear workers in Cambodia and Vietnam are at risk from automated assembly lines - or “sewbots” - according to a new [report](#) from the International Labour Organisation (ILO).

There are 9 million people, mostly young women, dependent upon jobs in textiles, garments, and footwear within the Association of Southeast Asian Nations (Asean) economic area, which includes [Cambodia](#), Indonesia, Thailand and Malaysia. These are the workers the ILO identifies as most

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Nine years have passed — have the garment workers of Bangladesh and Vietnam been replaced? No. Bangladesh alone has 5 million garment workers.

Over 50.17 lakh people working in garment sector in Bangladesh: state minister

Staff Correspondent 23 June, 2024, 21:39 [Best clothing retailers](#)



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There are over 50.17 lakh workers working in garment factories in the country and 55.57 per cent of them are women, state minister for labour and employment Nazrul Islam Chowdhury told Jatiya Sangsad on Sunday.

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I recently came across an article by Rodney Brooks, an MIT robotics professor. He is a Fellow of the Australian Academy of Science, former director of MIT's Computer Science and Artificial Intelligence Laboratory, and one of the founders of iRobot — decades of work in both academia and industry in robotics; definitely an insider.

In his article "Why Today's Humanoids Won't Learn Dexterity" he raises two important reasons:

1. Current humanoid learning is vision-based, but the dexterity of the human hand comes not only from vision but even more from touch. Robot systems trained without tactile data will find it very hard to achieve manual dexterity. The difficulty of this path is that tactile training data is far harder to obtain than visual data. Current laboratory research is still very preliminary, far from real commercialisation.

2. Current humanoids are limited in capability and cannot replace people from the outset, so they must work alongside people. But working with humans in the same workspace, in close proximity, is actually very dangerous. His advice: do not get too close to a full-size humanoid...

My advice to people is to not come closer than 3 meters to a full size walking robot. And the walking robot companies know this too. Even in their videos you will not see people close to a locomoting humanoid robot unless there is a big table between them, and even then the humanoids only shuffle around a little bit.

According to the estimates in his technical predictions post "Predictions Scorecard, January 1, 2025":

1) A dexterous robotic hand is impossible before 2030; he hopes it can be achieved by 2040.

Dexterous robot hands generally available.	NET 2030 BY 2040 (I hope!)	Despite some impressive lab demonstrations we have not actually seen any improvement in widely deployed robotic hands or end effectors in the last 40 years.
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2) A robot at the level of a cat or dog will not appear before 2048.

A robot that seems as intelligent, as attentive, and as faithful, as a dog.	NET 2048	This is so much harder than most people imagine it to be--many think we are already there; I say we are not at all there.
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3) A robot that understands humans like a six-year-old child will not be seen in his lifetime (before 2050).

A robot that has NIML
any real idea
about its own
existence, or the
existence of
humans in the
way that a six
year old
understands
humans.

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These predictions are constantly adjusted in line with the state of the industry, but one thing is certain: the humanoid robots currently eyeing commercialisation have no capacity to replace people at all, and whether they can truly enter human workplaces and assist human workers is a very big question mark.

Autonomous Driving

Autonomous driving is the most mature of AI applications; in many people's eyes, the driver's job will disappear within a few years. But industry experts do not see it that way.

Rodney Brooks' assessment is that autonomous driving cannot be fully commercialised before 2045. Today's so-called commercial self-driving cars either carry a safety operator inside or are controlled by humans in the background — they are not real autonomous driving.

The majority of US NET 2045
cities have the
majority of their
downtown under
such rules.

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The autonomous-driving hype has gone through several rounds; this is the latest one.

Some of the AI Hype Cycles that I Have Seen

- Pre-me:
 - Neural networks (vol 1), 1943, McCulloch & Pitts
 - Turing, 1950 computers fast enough for AGI
 - Neural networks (vol 2) 1950s, Minsky & Rosenblatt
 - McCarthy, 1956 computers fast enough for AGI
- Michie, 1959, reinforcement learning (1)
- Minsky, Michie, Nilsson, 1960: search
- It is all first order logic & resolution theorem proving
- Neural networks (vol 3) 1968 Minsky again
- MacHack (search, chess 1)
- Fuzzy logic, STRIPS, knowledge-based systems (revolution in medicine 1)
- Neural networks (vol 4) back propagation
- Primal sketch, 1979, Marr
- Self driving cars 1, 1987, Dickmanns
- 1980s reinforcement learning (2, & 3)
- SOAR, qualitative reasoning, support vector machines
- Self driving cars 2, 1997, Kanade et al
- Deep Blue (search, chess 2)
- Self driving cars 3, 2007, Thrun
- Bayesian inference
- Watson (Jeopardy & revolution in medicine 2)
- Deep learning, neural networks (vol 5)
- Self driving cars 4, 2010+, commercial
- Alpha Go, reinforcement learning (4)
- Generative AI, neural networks (vol 6)

Carmakers have been boasting of full autonomy for many years, just as Chinese carmakers boast of reaching L3 — when in fact L3 is achieved only in certain scenarios, and with a pile of qualifiers it amounts to no L3 at all.

FORECASTS: http://www.driverless-future.com/?page_id=384 March 27, 2017

- NVIDIA to introduce level-4 enabling system by 2018 (2017)
- NuTonomy to provide self-driving taxi services in Singapore by 2018, expand to 10 cities around world by 2020 (2016)
- Delphi and MobilEye to provide off-the-shelf self-driving system by 2019 (2016)
- Ford CEO announces fully autonomous vehicles for mobility services by 2021 (2016) ←
- Volkswagen expects first self driving cars on the market by 2019 (2016)
- GM: Autonomous cars could be deployed by 2020 or sooner (2016) ←
- BMW to launch autonomous iNext in 2021 (2016) ←
- Ford's head of product development: autonomous vehicle on the market by 2020 (2016) ←
- Baidu's Chief Scientist expects large number of self-driving cars on the road by 2019 (2016)
- First autonomous Toyota to be available in 2020 (2015) ←
- Elon Musk now expects first fully autonomous Tesla by 2018, approved by 2021 (2015)
- US Sec Trans: Driverless cars will be in use all over the world by 2025 (2015)
- Uber fleet to be driverless by 2030 (2015) ←
- Ford CEO expects fully autonomous cars by 2020 (2015) ←
- Next generation Audi A8 capable of fully autonomous driving in 2017 (2014)
- Jaguar and Land-Rover to provide fully autonomous cars by 2024 says Director of Research and Technology (2014)
- Fully autonomous vehicles could be ready by 2025, predicts Daimler chairman (2014) ←
- Nissan to provide fully autonomous vehicles by 2020 (2013) ←
- Truly autonomous cars to populate roads by 2028-2032 estimates insurance think tank executives (2013)
- Continental to make fully autonomous driving a reality by 2025 (2012)

Conclusion

With scaling laws hitting their ceiling, the limits of all other AI applications are settled as well. Whether robots or autonomous driving, achieving human-level dexterity requires not only perception but also reasoning. Reasoning is either rule-based or neural-network-based. Rule-based systems always miss the corners and edges; neural networks do not solve the problem of inaccuracy. In the digital domain of white-collar work, there may still be room to try. In the domain of physical labour, since personal safety is involved, how to promote the technology at all is a problem in itself. But without promotion, how to gather data for further training? The chicken and the egg — which comes first?

The development of AI, robots, autonomous driving, and similar technologies should, in principle, be a good thing for humanity. But if these technologies are used only for the ends of capital, they may bring disaster to mankind. Fortunately, technological development is not as formidable as Musk and company claim; humanity still has a chance to develop and apply AI under a better social system.

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